UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Wednesday 08 June 2023 09.00 to 12.00 (*3 hours plus any pre-agreed individual adjustment*)

Paper PHA/1

PHARMACOLOGY EXTENDED ANSWER (PAPER 1)

*All candidates must provide **one answer** for **each question** from the choice of a or b. You therefore need to provide **six** answers in total*

Diagrams or other handwritten material:
Where appropriate (e.g., for formulae, equations or diagrams) you may hand write your answers using the Inspera Scan Paper provided. You should not copy and paste images from any other documents.

Please Note:
- *Inspera Scan question codes MUST be noted during the exam.*
- *Please ensure that each Inspera Scan page is labelled appropriately, including your Blind Grade Number.*
- *Inspera Scan pages MUST be placed in the Inspera Scan folder at the completion of the exam and left on your desk.*
- *Inspera scan pages will be scanned by administrators after the completion of the exam.*

Stationery requirements:
- *Rough work pad*
- *Inspera Scan pages*
- *Inspera Scan Folder*

Calculator – students are permitted to use an approved calculator

1. **Either**

   a. Discuss the similarities and differences of neuromuscular blocking agents.

   or

   b. Compare and contrast the different pharmacological interventions for infertility.

2. **Either**

   a. Discuss the role of protein phosphorylation and other post-translational modifications in Growth Factor signalling.

   or

   b. The initial small molecule hits in a typical drug discovery initiative often need to be further optimised – discuss.

3. **Either**

   a. Compare and contrast the mechanisms of action of antiviral drugs.

   or

   b. Discuss the mechanisms of action of antimalarial drugs.

4. **Either**

   a. With reference to specific examples, discuss how the synthesis and signalling of lipid mediators derived from arachidonic acid can be targeted to treat disease.

   or

   b. Using specific examples, discuss the mechanisms of action of different immunosuppressive drugs.

5. **Either**

   a. Compare and contrast the therapies used to treat chronic heart failure.

   or

   b. Compare and contrast the mechanisms of action of drugs that target voltage-gated Na^+ channels and voltage-gated Ca^{2+} channels in treating diseases.

6. **Either**

   a. Compare the mechanisms of action of drugs targeting neurotransmitter reuptake transporters versus neurotransmitter receptors for treating neurological disorders.

   or

   b. Compare how drugs target serotonergic versus dopaminergic signalling to treat neurological disorders.

END OF PAPER